

Smart Power Distribution Board

The project is about the PCB design of a power distribution board. While we face some issues while distributing the power to different components in a project, this board deals with it.

It takes a single input and provides multiple output voltages (3.3V, 5V, 9V, 12V & one output for adjustable voltage) with basic circuit protection.

Features:

- Single input barrel jack
- Multiple regulated outputs
 - 3.3V
 - 5V
 - 9V
 - 12V
 - Adjustable output

Circuit Protection:

- Input reverse-polarity protection diode
- Resettable fuse (poly fuse) on input
- Individual output capacitors for each regulator

How to use ?

1) Connect input

- Plug a suitable DC adapter (e.g., 12 V) into the barrel jack
- Ensure adapter voltage and current ratings are within the regulator limits

2) Connect output

- Use screw terminals of the PCB to connect devices to 3.3 V, 5 V, 9 V, 12 V, or adjustable output
- Do not exceed the maximum current rating of each regulator or the polyfuse

Additional:

Add a trimpot (trimmer potentiometer) near LM317 if needed. I didn't get the correct footprint, in-case if needed you can place it in the design itself.